

Dual-Band Quad-Element mmWave MIMO Antenna with Parallel-Aligned Slots for 5G Applications

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Abstract—This study presents the design of a dual-band MIMO antenna for 5G mm-Wave communication, operating at 28 GHz and 32 GHz. The proposed antenna, measuring 20×20×0.8 mm³ and using a Rogers RT Duroid 5880 substrate, whose permittivity is 2.2, is incorporated rectangular slots to achieve dual resonance at 27.98 GHz and 32.02 GHz with return losses of -36.2 dB and -38 dB, respectively. The dual-band operation is achieved through an optimized microstrip patch design that incorporates two slots into the patch structure. The inset feed configuration is further enhanced the antenna's performance by providing better impedance matching and improved bandwidth characteristics. It is achieved peak gains of 7.84 dBi and 4.25 dBi at 27.98 GHz and 32.02 GHz. Additionally, the proposed dual-band MIMO antenna is demonstrated superior performance with an envelope correlation coefficient (ECC) of less than 0.000036, indicating minimal mutual interference between antenna elements. Total active reflection coefficient (TARC) is below -20 dB. The channel capacity loss (CCL) is maintained under 0.06 bits/sec/Hz, and the mean effective gain (MEG) is approximately -3.03 dB and -3.06 at respected resonant frequency. It achieves a diversity gain (DG) of approximately 9.99 dB and exhibited isolation between elements exceeding -22.5 dB.

Index Terms—MIMO, TARC, ECC, DG, CCL, MEG, Matlab, CST.

I. INTRODUCTION

In the contemporary age, the rapid growth of wireless devices, insufficient bandwidth, and constrained channel capacity have significantly speed up efforts to develop modern communication network standards. Since the popularity of modern communication devices has encouraged the development of fifth-generation (5G) communication systems, which utilizes the milli-meter wave spectrum and have substantially offered larger channel capacities and higher data speeds as well as low latency [1], [2]. with the implementation of 5G technology, several sector including virtual realities, the internet of things (IoT), self-driving cars, telecom sectors etc, are going to experience substantial changes [3]. Nonetheless, there are significant mm-wave spectrum limitations that need to be taken into consideration. The significance of these problems increases when a single element antenna [3] is being used. The multiple-input multiple-output (MIMO) antenna has been designated as a critical enabling innovation for recent wireless systems. It delivers the benefits of high data rates as well as throughput at gigabits per second, enhances channel capacity, and exhibits the simultaneous functioning of numerous antennas.

The ever-growing demand of MIMO antenna requires a broad bandwidth for concurrent functioning, a high gain to mitigate atmospheric diminutions and absorptions at millimeter wave frequencies, and a compact structure to facilitate assimilation in proposed MIMO system [1] which can be challenging in terms of physical deployment and isolation management. The integration of 5G technology with MIMO design enhanced those performance through multiplexing and spatial diversity. To reduce mutual coupling as well as maintain good isolation, a number of techniques have been examined. Here, edge to edge separation is utilized to enhance the isolation as well as reduced electromagnetic coupling of the antenna. It performs well to reduce coupling, but the front-to-back ratio might be negatively impacted by the emergence of back lobes, which should be considered at the time of improving isolation [4], [5]. In the area of 5G multiple-input multiple-output (MIMO) antenna technology, dual band also offer to resonate at two distinct mm-Wave frequency bands as well as wider available spectrum of the proposed antenna. Dual band can also be achieved by introducing extra components like slots, stubs, or parasitic patches or by modifying the patch's size and shaped [6]. Thus, the proposed dual band outperforms over single band antenna in terms of efficiency and adaptability [7]. Initially in 2021, Muhammad Kazim and his mate propounded a 4 port mimo antenna with poor mimo performance [8]. In same year, I.E Lamri and team [9] explore various performance indicators such as ECC, DG, CCL and suggested a small 2x2 UWB MIMO antenna. But, the isolation performance of both proposed antenna is comparatively poor. In recent year, 4 port mimo antenna was proposed to mitigate isolation performance [10]. The same year, Wahaj Abbas and all carried out a study and proposed a dual band mimo antenna with good isolation and channel capacity loss. In Spite of good isolation performance, Those are not operated at millimeter wave 5G application [8]–[11]. In 2022, a four-port mimo antenna is propounded that is functionated at millimeter wave frequency band. The performance of isolation enhance throughout the study [12]. But, no discussion is present regarding different mimo parameters such as ccl, tarc, meg, meg ratio etc [8], [10], [12]–[14].

This study introduces a dual-band MIMO structure with 4 elements that is intended for 5G mm-wave applications. In order to achieve dual band specifications in proposed

antenna, an effective and popular method is to use slots that provide the alternative current pathways through the patch of the antenna. The antenna's edge-to-edge distance among the MIMO elements was optimized to enhance performance for Mutual coupling, ECC, isolation performance, DG etc. By evaluating key MIMO parameters such as TARC, MEG, MEG Ratio, CCL, and ECC. Our goal is to develop a stringent and effective antenna system that can satisfy the demanding performance standards of 5G mm-wave application.

The next section is organized as follows: Proposed MIMO antenna design procedure in section II, Working principle of proposed antenna are in section III, Section IV are in simulated experimental result and discussion. At last, Section V covers conclusion of the work.

II. ANTENNA DESIGN PROCEDURE

This section discloses the entire design of our suggested MIMO antenna for 5G milli-meter wave application in order to achieve improved performance. This section has been split down into two subsections to facilitate understanding: Initially, a single element antenna was developed to achieve dual resonance within the antenna element. Then, 4 identical elements are placed with a specific separation distance to obtain MIMO structure.

A. Proposed Antenna's Specifications

The proposed antenna is placed on a 0.8 mm thickness Rogers RT Duroid 5880 substrate layer that exhibits a relative dielectric constant (ϵ) of 2.20 above the copper-annealed ground layer. The dimension of the proposed single element ground is $8 \times 8 \text{ mm}^2$ shown in Fig. 1. The single antenna is equipped with a rectangular microstrip patch antenna, two rectangular slots and a 50Ω microstrip line. .

TABLE I: Comprehensive overview of the characteristics and selected values of the chosen patch antenna

Design Specifications	Measures
Dielectric Component	Rogers RT Duroid 5880
Dielectric Constant	2.20
Tangent Loss ($\tan\delta$)	0.02
Substrate thickness (h)	0.80 mm
Conductor thickness (Copper)	0.035 mm

B. MIMO Antenna's Configuration

The suggested MIMO design is formed in 4 identical elements patch antenna arranged in front of each other, as depicted in Fig. 2. The overall dimension of the proposed MIMO design are $20 \times 20 \times 0.8 \text{ mm}^3$. The single element of the MIMO antenna's parameters were modified after-optimization, with the final parameters presented in Table II. The MIMO antenna configuration thus achieved decent performance, but further improvement and reduction of mutual coupling among the antennas are needed. To enhance isolation as well as mutual coupling performance, the distance between the antenna elements edge must be increased uniformly so that it can minimize the front-to-back ratio of the MIMO configuration.

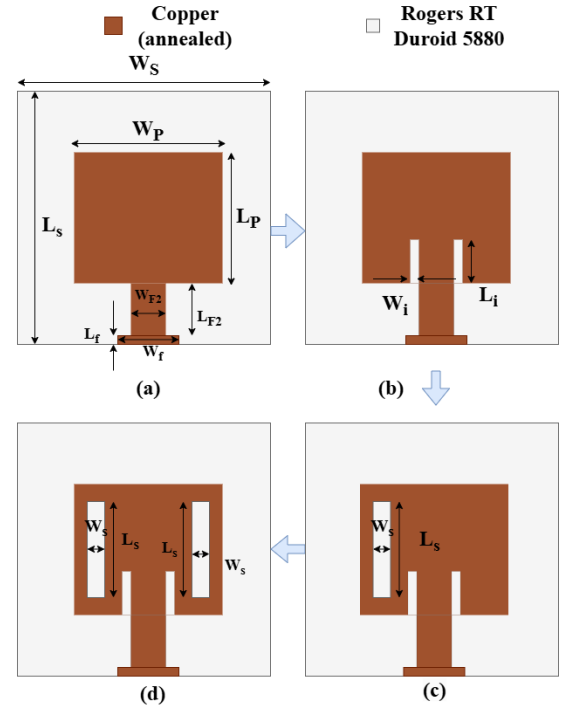


Fig. 1: Proposed single patch design procedure (a) Rectangular Patch (b) Inset feed patch antenna (c) Proposed single patch element with single slot (d) Proposed single patch element with dual slot.

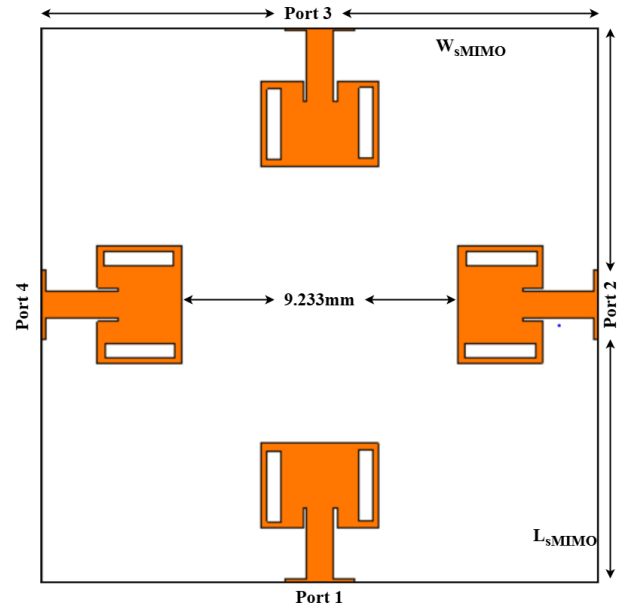


Fig. 2: Proposed symmetric orientation MIMO antenna.

III. WORKING PRINCIPLE OF PROPOSED ANTENNA

The Dual-Band Quad-Element mmWave MIMO antenna with parallel-aligned slots operates by utilizing carefully designed parallel slots that generate dual-band functionality, typically covering both 27.98 GHz and 32.02 GHz that

TABLE II: Designing parameters corresponding to the Proposed MIMO antenna

Design Parameters	Values(mm)	Design Parameters	Values(mm)
L_{sMIMO}	20	W_{sMIMO}	20
L_p	3.063	W_p	4.236
W_f	2.415	L_f	0.15
L_{f2}	1.71	W_{f2}	0.95
L_i	0.73	W_i	0.08
W_s	0.51	L_s	2.54

are occupied in mm-wave frequency range. The dual-band characteristics are achieved by incorporating two rectangular slots into the proposed patch, which create alternate current pathways, enabling resonance at 27.98 GHz and 32.02 GHz. To obtain proper impedance matching between the feed line and the antenna, The feed line is simply inserted into the center of the antenna element that is shown as Fig. 1(b). These slots are engineered to resonate at two distinct frequencies, enabling simultaneous operation at both bands. Each of the four antenna elements operates in parallel, enabling spatial diversity and improving signal quality by reducing interference and increasing capacity in a 5G network, making it suitable for high-performance 5G mm-wave applications.

IV. SIMULATED RESULTS AND DISCUSSIONS

A. Optimization of Single element Length and Width

A parametric analysis was carried out on the single element patch to optimize the performance. The main focus was to enhance the return loss of the single element that is exhibited in Fig. 3.

Based on the return loss, the single element was approved to design the proposed 4-port MIMO antenna. It offers optimum performance in slot length and width at $L_s = 2.54$ mm and $W_s = 0.51$ mm respectively.

B. Reflection coefficient of MIMO Antenna

The MIMO antenna's reflection coefficient S_{11} is demonstrated in Fig. 4. From this Figure, S_{11} , S_{22} , S_{33} and S_{44} parameters were gathered. The modified result of S_{11} for proposed single elements are also depicted in Fig. 4. The lowest possible value of S_{11} for the suggested MIMO antenna is roughly -36.20 dB at 27.98 GHz and -38 dB at 32.02 GHz respectively.

C. Isolation performance

The ability of each antenna element to transmit a signal without interfering with signals coming from other antenna components is known as isolation performance. The S_{12} , S_{13} , S_{14} , S_{32} , S_{42} and S_{43} plot pertaining to the resonance frequency of our suggested antenna was discovered, as demonstrated in Fig. 5. Our suggested MIMO antenna has isolation of less than -20 dB, or roughly -22.5 dB, based on the results demonstrated in Fig. 5, which denotes a high degree of isolation between antenna parts and little signal interference from one antenna to the others.

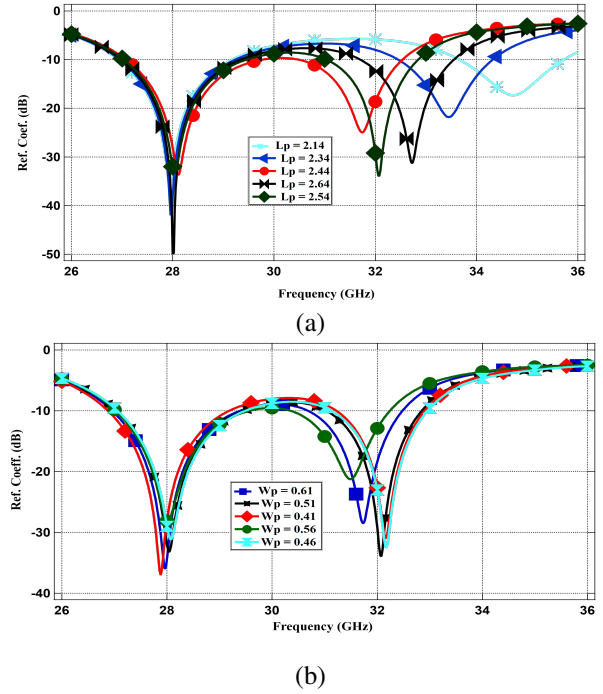


Fig. 3: Reflection coefficient of a single element antenna: (a) for different slot lengths, and (b) for different slot widths.

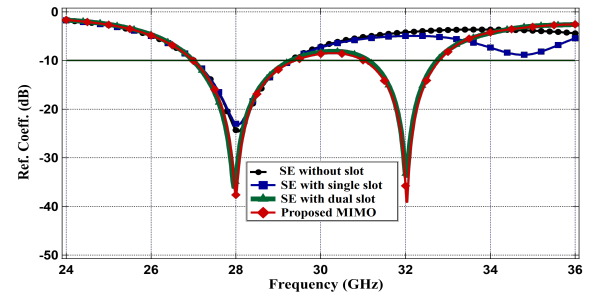


Fig. 4: Simulated reflection coefficient of the single element (SE) and MIMO antenna.

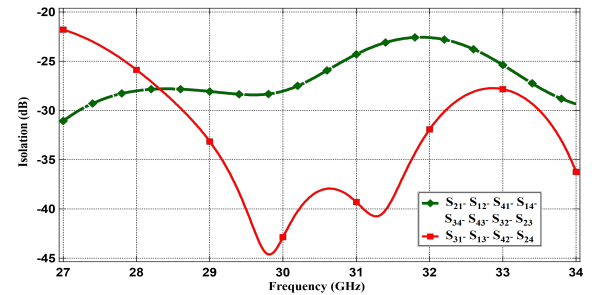


Fig. 5: Isolation performance of the MIMO antenna.

D. Gain of the proposed MIMO

Fig. 6 depicts the gain achieved by the recommended antenna. At 27.98 GHz resonant frequency, The peak gain is found in 7.84 GHz. On the other hand, the gain is around 4.24GHz at 32.02GHz. The radiation efficiency is more than 83% entire the operating frequency.

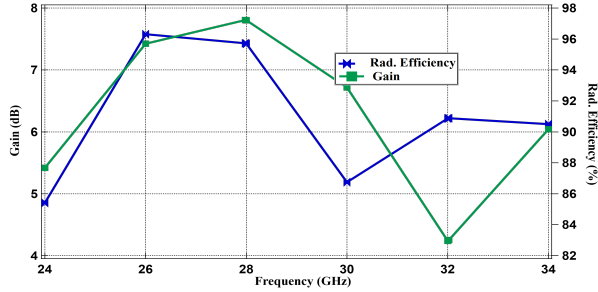


Fig. 6: Gain of the suggested mimo antenna.

E. Far Field Radiation Pattern

Fig. 7 and 8 exhibits the radiation patterns of the reference antenna in the two-dimensional at 27.98 GHz and 32.02 GHz and it is evident that the proposed antenna is radiating in a broadside orientation for both frequencies. This orientation also illustrates how the antenna may radiate signal power distribution in vertical directions. It enables the antenna to adjust to different circumstances and signal conditions, which reduces the probability of signal dropouts and enhances connection in intricate situations involving multipath effects, obstructions, or reflections of signals. These characteristics guarantee the antenna's excellent efficacy for 5G applications involving high frequency millimeter waves.

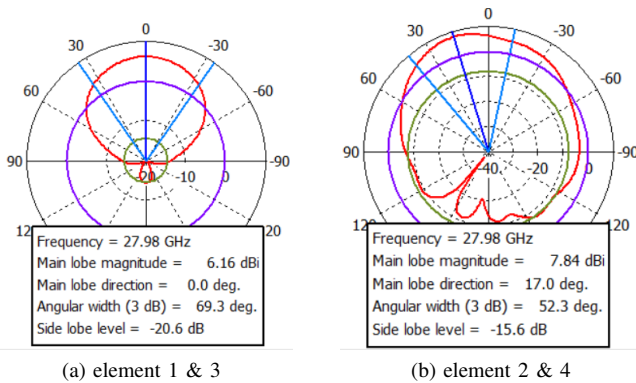


Fig. 7: 2D radiation pattern for our designed MIMO in (a) element 1 & 3 (b) element 2 & 4 at 27.98 GHz frequency.

Another crucial factor to consider while evaluating an antenna's performance is its radiation pattern. The radiation pattern at 27.98 GHz and 32.02 GHz is given in fig. 9 and 10, respectively. It can be shown that the recommended work gives directional radiation in the E-plane and H-plane at 27.98 GHz and 32.02 GHz. Co-polarization is typically

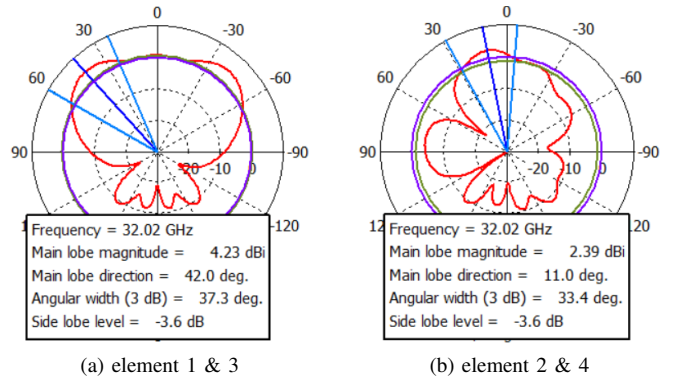


Fig. 8: 2D radiation pattern for our designed MIMO antenna in (a) element 1 & 3 (b) element 2 & 4 at 32.02 GHz frequency.

stronger, indicating that the antenna is designed to focus on a specific polarization.

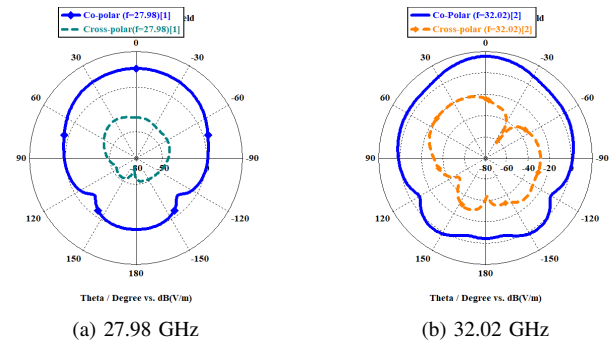


Fig. 9: Simulated co- and cross polarization radiation patterns for E-plane of our designed MIMO antenna.

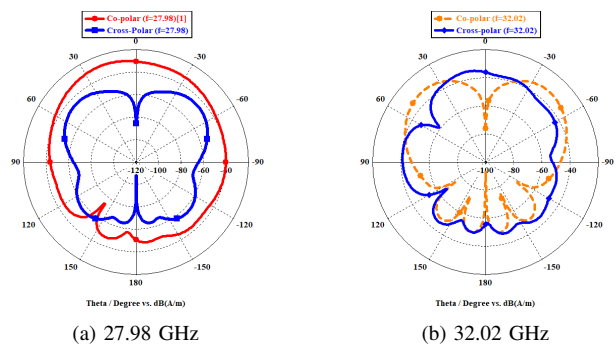


Fig. 10: Simulated co- and cross polarization radiation patterns for H-plane of our designed MIMO antenna.

F. Envelope correlation Coefficient (ECC)

The ECC is crucial in analyzing the performance of the MIMO antenna. As exhibited in Fig. 11, the ECC value at 27.98 GHz is determined to be around 0.000036. According

to these ECC values, the suggested antenna exhibited reduced mutual coupling and increased diversity.

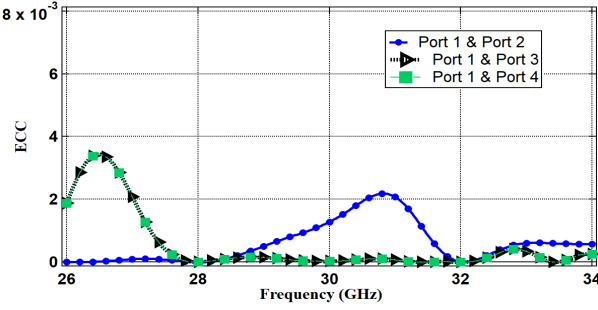


Fig. 11: Envelop correlation coefficient performance (ECC) of the proposed MIMO Design.

G. Diversity Gain

Diversity gain (DG) illustrates the transmission power loss that occurs when diversity techniques are utilized to the MIMO configuration. 10 dB is the optimal DG value. The DG of the MIMO antenna is realized in Fig. 12 and is approximately 9.99 dB, indicating a good SNR enhancement of the MIMO design.

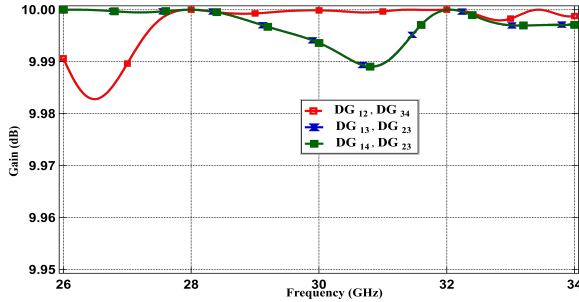


Fig. 12: Diversity gain of proposed MIMO structure.

H. Channel capacity loss (CCL)

Channel capacity loss (CCL) is a suitable MIMO parameter for explaining a MIMO antenna's performance. The CCL is equivalent to a value less than 0.06 bit/sec/Hz within the working operating frequency bands, which is the ordinary value and is also accepted range in practice.

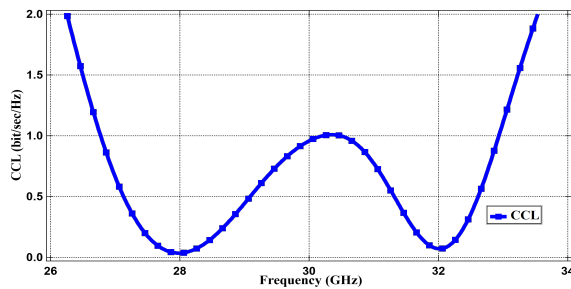


Fig. 13: CCL of the suggested MIMO antenna.

I. Total active reflection coefficient (TARC)

The amount of power that is reflected back from a system or material relative to the amount of power incident upon it is measured by the total active reflection coefficient, or TARC. The TARC of a MIMO antenna system should be less than -10 dB for better communication [14]. Fig. 14 shows that the TARC of a MIMO antenna system should be less than -10 dB. TARC is calculated when the input signals of other three elements are energized with a phase differences between 0° to 120° while keeping the first port constant. A phase shift of about -20 dB at 90° across the whole bandwidth. Consequently, any TARC value. The phase differences between the excitation signals at different ports significantly influence TARC. Constructive or destructive interference of reflected waves, governed by these phase relationships, determines the overall reflected power.

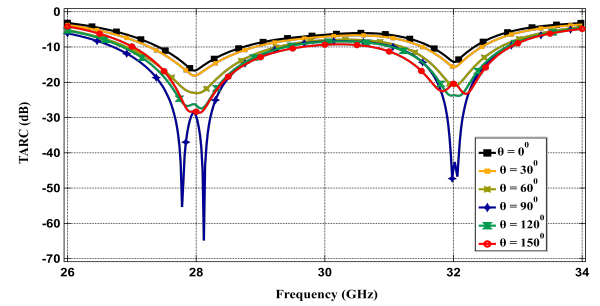


Fig. 14: Simulated total active reflection coefficient (TARC) of the suggested MIMO antenna

J. MEG and MEG Ratio of the proposed MIMO

The S-parameters and the antenna system's spatial correlation can be used to calculate the Mean Effective Gain (MEG) for a 4-port MIMO system. The MEG value should be in between $-3 \text{ dB} \leq \text{value} \leq -12 \text{ dB}$. As depicted in Fig. 15 the MEG value is approximately -3.06 dB to -3.6 dB throughout the functional band of the proposed MIMO design. From Fig. 15 (b) the various graph ratio of MEG-1 to MEG-2, MEG-3 and MEG-4, MEG-2 to MEG-3 and MEG-4 and MEG-3 to MEG-4 are approximately equal to unity, in turn indicating the quality of the proposed design.

K. Comparison

An overview of the suggested MIMO antenna and a comparison with previous mimo antenna and their parameter in this field's literature are displayed in Table III. The table provides a summary of the proposed MIMO antenna and compares it to existing MIMO antennas. It highlights key parameters such as ECC, Isolation, DG, CCL in the field's literature.

CONCLUSION

This study describes the successful development and characterization of a dual-band MIMO antenna operating at 27.98 GHz and 32.02 GHz for 5G mm-wave applications. The

TABLE III: Comparative Analysis of MIMO design with Pervious works

Reference	Dimension of patch (mm ²)	No of Port	ECC	Diversity Gain (dB)	Isolation (dB)	Peak Gain (dBi)	CCL (Bits/sec/Hz)	TARC (dB)
[8]	60 × 60	4	≤ 0.022	9.90	< −15	—	—	—
[9]	65 × 65	4	≤ 0.025	9.99	< −15	—	0.08	—
[10]	60 × 68	4	≤ 0.001325	9.99	< −20	< 4.5	—	—
[11]	—	2	4×10^{-1}	9.99	< −22	6.25	0.1	—
[12]	20 × 20	4	≤ 0.0013	—	< −18	9.24	—	—
[13]	15 × 15	4	≤ 0.001	—	< −18	—	0.5	−10
Proposed	20 × 20	4	≤ 0.000036	9.98	< −22.5	7.84	0.05	−20

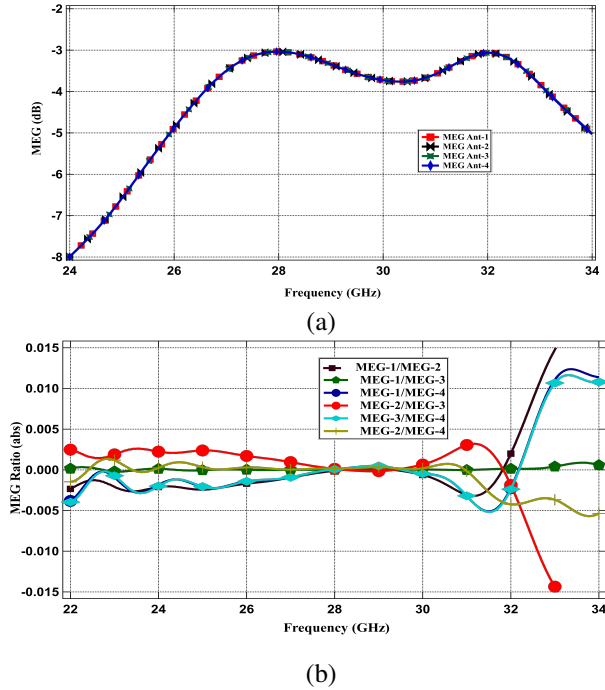


Fig. 15: The proposed MIMO antenna: (a) Mean Effective Gain (MEG), and (b) Mean Effective Gain Ratio (MEGR).

antenna was developed to fulfill subsequent-generation wireless communication systems strict specifications for high data throughput, low latency, and broad coverage. So, the proposed antenna design integrates suitable isolation techniques to reduce interference among the numerous antenna elements to achieve a high level of isolation, reduced ECC and around 10 dB diversity gain improve system performance by eliminating mutual coupling. Furthermore, The TARC and CCL performances of MIMO are promising. The results of the simulation and observations indicate how effectively the proposed antenna performs in achieving the desired performance criteria for mmWave 5G Application.

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